

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	
Project Title	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to predict the Human Development Index (HDI) of a country using Machine Learning. It analyzes important socio-economic indicators such as **Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and Gross National Income (GNI)** to provide fast and accurate HDI predictions. This helps researchers, students, and government analysts make better decisions without performing complex manual calculations.

Project Overview	
Objective	To develop a machine learning-based web application that predicts the Human Development Index (HDI) quickly and accurately using socio-economic indicators.
Scope	The system predicts HDI using four key development indicators and provides the predicted HDI value along with its development category through a simple web interface.
Problem Statement	
Description	Calculating HDI manually requires collecting multiple development indicators and performing several calculations, which is time-consuming and may lead to errors.

Impact	The proposed system reduces manual HDI calculations and provides quick, accurate predictions, helping researchers, students, and policy makers analyze development indicators efficiently.
Proposed Solution	
Approach	Use a trained Linear Regression model to predict the Human Development Index (HDI) from four socio-economic indicators entered by the user.
Key Features	• Predicts HDI instantly. • User-friendly web interface. • Displays HDI Score and Development Category. • Reduces manual calculations. • Uses Linear Regression for prediction.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i3 or above
Memory	RAM	Minimum 4 GB
Storage	Disk space	Minimum 5 GB Free Space
Software		
Frameworks	Backend frameworks	Flask
Libraries	ML libraries	scikit-learn, pandas, numpy, joblib

Development Environment	IDE	VS Code / Jupyter Notebook
Data		
Data	Data Source	UNDP Human Development Dataset